By 2nd Distributive law, $\exists x(((\neg P(x) \vee \neg Q(x)) \wedge P(x)) \vee ((\neg P(x) \vee \neg Q(x)) \wedge Q(x)))$. By 2nd Commutative law, $\exists x((P(x) \wedge (\neg P(x) \vee \neg Q(x))) \vee (Q(x) \wedge (\neg P(x) \vee \neg Q(x))))$. By 2nd Distributive law, $\exists x(((P(x) \wedge \neg P(x)) \vee (P(x) \wedge \neg Q(x))) \vee ((Q(x) \wedge \neg P(x)) \vee (Q(x) \wedge \neg Q(x))))$. By Negation law, $\exists x((F \vee (P(x) \wedge \neg Q(x))) \vee ((Q(x) \wedge \neg P(x)) \vee F))$. By 1st Commutative law, $\exists x(((P(x) \wedge \neg Q(x)) \vee F) \vee ((Q(x) \wedge \neg P(x)) \vee F))$. By 2nd identity law, $\exists x((P(x) \wedge \neg Q(x)) \vee (Q(x) \wedge \neg P(x)))$ is True.

$\therefore$ There exists an element a in the domain s.t. $(P(a) \wedge \neg Q(a)) \vee (Q(a) \wedge \neg P(a))$ is True. By defn. of OR, the following cases arise:

① Only $P(a) \wedge \neg Q(a)$ is True. By defn. of AND, $P(a)$ is True and $\neg Q(a)$ is True. By defn. of negation, $\neg(\neg Q(a))$ is False. By double-negation law, $Q(a)$ is False. By defn. of universal quantifier, $\forall x Q(x)$ is False. We cannot conclude anything on the truth value of $\forall x P(x)$. $\therefore$ The truth value of $\forall x P(x) \leftrightarrow \forall x Q(x)$ is not determined, not necessarily False.

$\boxed{\therefore \text{They are not logically equivalent}}$ □

47) Show that $\exists x(P(x) \vee Q(x))$ and $\exists x P(x) \vee \exists x Q(x)$ are logically equivalent.

Soln: Let $\exists x(P(x) \vee Q(x))$ be True. By defn. of existential quantifier, there exists atleast one element in the domain such that the expression is True. Let a be such an element such that $P(a) \vee Q(a)$ is True. By defn. of OR, the following 3 cases arise: ① $P(a)$ and $Q(a)$ are both True. $\therefore$ By defn. of existential quantifier, $\exists x P(x)$ is True and $\exists x Q(x)$ is True. By defn. of OR, $\exists x P(x) \vee \exists x Q(x)$ is True.